SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

SignFlow Viva Preparation Guide

Simple answers for project questions and theory questions

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How to use this guide

Start with the one-minute project explanation. Then practise the project questions. For theory, learn the bold word first and then its one or two sentence answer. Use SignFlow examples wherever possible.

Important for the viva: SignFlow is a proposed project design. Say *the system will* or *the proposed design does*, not that the app has been deployed.

1. Your one-minute project introduction

You can say this almost exactly:

SignFlow is a proposed digital file approval system for a university. Today, approval files move physically between offices. This creates delays, lost files, and confusion about who has the file. The average approval time is around 12 days.

Our solution is a digital workflow. A staff member submits a file online, the system gives it a unique ID, and sends it to the correct officers based on file type and amount. Officers can approve, reject, or return the file. The system also keeps a full history, sends reminders for overdue work, and shows reports.

We recommend starting with purchase requests, leave applications, and event approvals. These three cover 1,600 out of 2,000 monthly files, which is 80 percent. The target is to reduce approval time to under 4 days and make sure no submitted file is lost.

Quick facts to remember

Point	Answer
Main problem	Paper files are slow, hard to track, and can be lost.
Monthly volume	2,000 files.
Current average time	Around 12 days.
Target	Under 4 days and zero lost submitted files.
First release	Purchase requests, leave applications, and event approvals.
Why these three?	They cover 1,600 files per month, or 80 percent of the total.
Estimated effort	27 person-days for three types; 108 person-days for all 12 types.

Person-day means the work one person can do in one working day. It is an effort estimate, not automatically the calendar duration.

2. SignFlow project questions

What problem does SignFlow solve?

It solves slow paper-based approval. A paper file may be delayed, lost, or difficult to locate. SignFlow gives every file a digital record, shows who must act next, and keeps a history of actions.

Who uses the system?

Administration staff create and submit files. Section officers review them. Senior officers approve and may sign them. The Registrar monitors outcomes. The IT team manages the technical side. Legal or compliance staff approve the digital-signature method.

What happens after a file is submitted?

The system checks required details and attachments, creates a unique file ID, chooses the approval route, sends it to the first officer, records the action in audit history, and shows the current owner and due date.

What is routing?

Routing means choosing who receives a file next. For example, a high-value purchase request may need a section officer, a senior officer, and then an authorised approver.

Why not digitise all 12 file types at once?

Because all 12 types need more rules, testing, and change at the same time. Starting with three high-volume types gives 80 percent coverage with less risk. The team can learn from the pilot before expanding.

What is an audit trail?

An audit trail is a record of actions: who created a file, who approved it, when it was returned, who signed it, and when reminders were sent. It improves accountability.

What are the main functional requirements?

The system must create files, validate fields, route files, allow approval/rejection/return, handle signatures, send reminders, keep audit history, support search, and provide reports.

What are the main non-functional requirements?

The system should be secure, fast, available, auditable, and reliable. Examples are 99.5 percent availability, 95 percent of normal pages within 3 seconds, and zero lost submitted files.

3. SignFlow design, UML, testing and risk

What is HLD and LLD?

HLD means High-Level Design. It shows the big picture: major modules and how they connect. LLD means Low-Level Design. It gives detailed design such as classes, fields, methods, and logic.

What is the high-level design of SignFlow?

The main modules are file intake, routing, approval, digital signature, notifications, audit history, search, and reporting. The user uses a browser interface, while the system may connect to university login, notification, and signature services.

Why are routing and signature modules separate?

They do different jobs. Routing decides who should approve a file. The signature module checks whether a signature is valid. Keeping them separate makes future changes easier, for example changing the signature provider.

Explain cohesion and coupling using SignFlow.

Cohesion means one module does one related job. For example, the Signature Service handles signature requests and validation only. Coupling means how much modules depend on each other. Low coupling is good: the Routing Service does not need signature-provider details.

Explain the use-case diagram.

It shows the users and what they do. Administration staff create files. Officers review and decide. Senior officers may sign. The Registrar views status and reports. Legal staff approve the signature method.

Explain the class diagram.

It shows important objects. A File has ID, type, amount and status. An ApprovalRoute contains ApprovalSteps. A DigitalSignature stores signing information. An AuditEvent stores who did what and when.

Explain the sequence diagram.

It shows the order of messages when a user submits a file: user to user interface, interface to file service, file service checks routing rules, the route is created, audit is recorded, first approver is notified, and the user sees file ID and status.

Explain the activity diagram.

It shows the workflow with decisions. The system checks whether required data is valid. If not, it shows errors. If yes, it routes the file. Later, the officer may approve, return, or reject it. Overdue files trigger reminders or escalation.

Explain the state diagram.

It shows a file changing state: Draft, Submitted, In Review, Awaiting Signature, Approved, Returned, Rejected, Cancelled, and Archived.

What are the major risks?

Senior officers may resist digital work; signatures may not receive legal approval; wrong rules may route files to the wrong person; and rules for remaining file types may be incomplete.

How will you reduce those risks?

Use training and assisted support, get legal approval before release, test routing boundaries carefully, get business sign-off on routes, and begin with only the three best-understood file types.

4. Module I: Introduction to Software Engineering

What is software engineering?

Software engineering is a systematic way to plan, build, test, maintain, and manage software. It is not only coding. It also includes requirements, design, testing, documents, and project management.

What is the nature of software?

Software is intangible, meaning we cannot physically touch it like a machine. It is complex because many parts work together. It changes often when user needs change. Software does not wear out physically like hardware, but it can become difficult to maintain after many uncontrolled changes.

What is the software crisis?

When software systems became bigger, many projects were late, costly, full of bugs, and hard to maintain. This situation was called the software crisis. Software engineering introduced proper methods to reduce these problems.

What are important software engineering principles?

Understand the problem before coding; divide a large system into smaller modules; keep modules simple; plan for change; test continuously; document important decisions; and reuse reliable components where suitable.

What is SDLC?

SDLC means Software Development Life Cycle. It is the full path of software work: planning, requirements, design, development, testing, deployment, and maintenance.

Software process models

Model	Simple meaning	When useful
Waterfall	Finish one phase before the next: requirements, design, build, test.	When requirements are clear and stable.
Incremental	Build the system in small useful parts.	When value can be delivered step by step. SignFlow uses this idea: first three file types, then later types.
Spiral	Repeat planning, risk analysis, development, and evaluation in cycles.	For large or risky projects.
Prototype	Create an early model or screen to get feedback.	When users need help visualising the system. SignFlow UI screens are prototypes.
Agile	Build in short cycles, get feedback, improve often.	When needs can change and regular feedback is useful.

Why is the Incremental model suitable for SignFlow?

The approval rules for all 12 file types may not be ready. An Incremental approach allows the team to deliver the most important three types first, get feedback, reduce risk, and add remaining types later.

5. Module II: Requirements Engineering

What is requirements engineering?

It is the process of finding, understanding, documenting, checking, and managing what users need from a system.

What are the main steps?

Elicitation: collect requirements. Analysis: find conflicts or missing details. Specification: write requirements clearly. Validation: check that they are correct and testable. Management: control future changes.

What is requirement elicitation?

Requirement elicitation means collecting needs from stakeholders. Common methods are interviews, questionnaires, observation, workshops, document study, and prototypes.

Which elicitation techniques fit SignFlow?

Interview the Registrar and IT head; use questionnaires with administration staff and section officers; and walk through prototype screens with senior officers.

What is a functional requirement?

It states what a system must do. Example: SignFlow shall route a submitted file using configured file type and amount rules.

What is a non-functional requirement?

It states how well the system should work or a constraint it must follow. Example: 95 percent of normal pages should respond within 3 seconds.

What is requirement analysis?

It means checking requirements for ambiguity, missing information, conflicts, feasibility, and priority. For example, the desire to digitise all types immediately conflicts with limited effort and incomplete rules.

What is an SRS?

SRS means Software Requirements Specification. It is a formal document that explains system purpose, scope, users, functional and non-functional requirements, interfaces, assumptions, and acceptance criteria.

What is a feasibility study?

It checks whether a project is practical. Technical feasibility asks can we build it. Economic feasibility asks is the cost reasonable. Operational feasibility asks will people use it. Legal feasibility asks is it allowed. Schedule feasibility asks can it be finished on time.

What is requirement validation?

It checks that requirements are correct, complete, consistent, realistic, and testable. The requirement 'approval time under 4 days' is testable because it can be measured.

What is requirement management?

It tracks requirements and controls changes. If the university later asks for circular approvals, it should be recorded, analysed, approved, and planned rather than added without control.

6. Module III: Software Design and UML

What are design concepts?

Design concepts are ideas used to make a system easy to understand and change. Examples are abstraction, modularity, separation of concerns, cohesion, coupling, and information hiding.

What is abstraction?

Abstraction means focusing on important details and hiding unnecessary detail. For example, an approver only needs to see a file, action buttons, due date, and status. They do not need to know internal database details.

What is modular design?

It means dividing one large system into smaller parts, called modules. Benefits are easier understanding, testing, maintenance, and reuse.

What is architectural design?

It is the high-level structure of the system. It identifies major modules, interfaces, data flow, and external systems.

What is object-oriented design?

It designs software around objects. An object has data and behaviour. Example: a File object has file ID and status; it can be submitted or have its status changed.

UML quick answers

Diagram	What it shows	SignFlow example
Use case	Users and their goals.	Staff creates a file; officer approves; Registrar sees reports.
Class	Objects, data, methods, relationships.	File, StaffUser, ApprovalRoute, AuditEvent.
Sequence	Messages in time order.	Submit file -> choose route -> record audit -> notify officer.
Activity	Step-by-step work flow and decisions.	Validate file; route it; approve/return/reject it.
State	Different states of one object.	Draft -> Submitted -> In Review -> Approved.
Deployment	Where software parts run.	Browser, application server, database, signature service.

Easy memory line: use case = who does what; class = what objects exist; sequence = who talks first; activity = what happens next; state = what condition the object is in.

7. Module IV: Testing and Quality Assurance

What is verification and validation?

Verification asks: are we building the product correctly? Validation asks: are we building the correct product for users? Example: verifying SignFlow routing matches approved rules; validating that staff find the workflow useful.

What is unit testing?

Testing one small unit by itself. Example: check that validation rejects a file with no file type.

What is integration testing?

Testing whether modules work together. Example: after submission, check that routing, audit recording, and notification all work together.

What is system testing?

Testing the complete system as one whole system. Example: submit a purchase request and follow it through final approval.

What is acceptance testing?

Testing with users or customer representatives to decide whether the system is ready to accept. SignFlow acceptance includes correct routing, complete audit history, signature approval, and no lost test files.

What is white-box testing?

Testing internal code logic, conditions, branches, and loops. The tester knows the internal code.

What is black-box testing?

Testing based on input and expected output without looking at the internal code. Example: submit an invalid file type and check that the expected error appears.

What is regression testing?

Re-testing old features after a change. If purchase-request routing changes, test leave and event approval routes too, so they are not broken.

What is Boundary Value Analysis?

It tests values at the edge of an allowed range. If reminder interval is 1 to 30 days, test 0, 1, 2, 29, 30, and 31.

What is equivalence partitioning?

It divides inputs into groups that should behave the same. For reminder interval, valid group: 1 to 30. Invalid groups: below 1, above 30, text, or decimals.

What is Software Quality Assurance?

SQA is the planned way to make sure software meets required quality. It includes standards, reviews, tests, audits, defect tracking, and process improvement.

What is DRE?

DRE means Defect Removal Efficiency. Formula: defects found before release divided by total defects found before and after release, multiplied by 100. A high value means more defects were caught before users saw them.

What is software maintenance?

Maintenance means changing software after release. Corrective fixes bugs. Adaptive changes it for a new rule or environment. Perfective improves performance or usability. Preventive reduces possible future problems.

8. Module V: Software Project Management

What is project management?

It is planning, organising, monitoring, and controlling work so that a project meets its scope, time, cost, and quality goals.

What is a WBS?

WBS means Work Breakdown Structure. It breaks a large project into smaller manageable tasks. SignFlow WBS includes initiation, requirements, design, build/configuration, testing, training, pilot, and closure.

What is a Gantt chart?

A Gantt chart is a timeline showing tasks, start and finish dates, duration, and sometimes dependencies.

What is PERT?

PERT means Program Evaluation and Review Technique. It helps estimate uncertain task duration. Formula: Expected time = (optimistic + 4 times most likely + pessimistic) / 6.

What is CPM?

CPM means Critical Path Method. It finds the longest chain of dependent tasks. A delay on this chain delays the whole project.

What is the critical path in SignFlow?

Initiation -> Requirements -> Design -> Configuration and Build -> Integration Testing -> UAT and Training -> Pilot and Closure. The planning estimate is 32 working days.

What is cost estimation?

It predicts the effort, time, and money needed. For SignFlow we use supplied person-day estimates because there is no reliable code-size or cost rate in the case study.

What is COCOMO?

COCOMO means Constructive Cost Model. It estimates effort using software size, often in KLOC, which means thousand lines of code. Basic form: Effort = a times KLOC raised to b. We do not calculate it for SignFlow because reliable code-size data is not given.

What is risk management?

It means identifying future problems, estimating their chance and impact, planning responses, and monitoring them. A risk may happen in the future; an issue has already happened.

What is configuration management?

It controls versions of project items such as SRS, design, code, test cases, and documents. It makes sure the team knows which version is correct.

What is Git?

Git is a version-control tool. A repository stores the tracked project. A commit is a saved checkpoint. A branch is a separate line of work. A merge combines changes.

What is Agile project management?

It manages work in short cycles with regular feedback. The team prioritises useful work, delivers a small improvement, reviews it, and repeats.

What is Scrum?

Scrum is an Agile framework. Main roles: Product Owner decides priorities; Scrum Master supports the process; Development Team builds the work. Work happens in short sprints.

What is a sprint and sprint planning?

A sprint is a fixed short work period, often one to four weeks. Sprint planning is the meeting where the team chooses what it will complete in the coming sprint.

9. Tough questions: ready-made answers

Why did you choose Incremental development?

The university has 12 file types, but all rules may not be clear at the start. Incremental development lets us implement the three highest-volume types first, receive feedback, reduce risk, and expand later.

Why not Waterfall?

Waterfall works best when requirements are stable. In this project, approval rules, legal signature approval, and senior-officer usability may change after feedback. So an Incremental approach with prototypes is more suitable.

What is the difference between risk and issue?

A risk is a future possibility. Example: the signature provider may not get approval. An issue is a problem that already exists. Example: paper files are already being lost or delayed.

How will you know the project is successful?

We will measure average approval time under 4 days, zero lost submitted files, correct routing, complete audit history, legal approval for digital signatures, and successful user acceptance testing.

What if signature validation fails?

The file must not become finally approved. It should stay in an exception or review state until the problem is resolved. This prevents an invalid signature from being accepted.

What if a file is sent to the wrong officer?

Pause or correct the affected routing rule, test the corrected route, get approval from the business owner and IT lead, and then activate it again.

What if an officer rejects a file?

The system records the rejection, officer identity, time, and reason. The requester can read the reason and correct the file if appropriate.

Last-minute speaking tips

- Answer in two or three sentences first. Add detail only if the examiner asks.
- Use SignFlow examples: file ID, routing, due date, reminder, audit trail, signature.
- Do not claim that the proposed app is already deployed or fully built.
- If a value is an assumption, say: 'This would be confirmed during requirements validation.'
- If you forget a definition, explain it in plain language with a project example.

Final line to remember

The main idea of SignFlow is simple: every approval file should be easy to find, sent to the right person, completed faster, and recorded properly.

Prepared for Sourabh Yadav | Software Engineering and Project Management viva